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# Depth Camera-based Monitoring and Simulation System for Weight Training Assessment

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***Abstract*— This paper introduces a depth camera-based monitoring and simulation system, specifically designed for assessing weight training. Recognizing the limitations of Internet of Thing (IoT) devices and wearable sensors in providing direct exercise guidance, this study employs a depth camera for real-time posture monitoring. The system integrates skeleton recognition to capture joint trajectories, feeding them into an OpenSim-based upper limb model for detailed muscle and joint analysis. This approach allows assessment of muscle fiber and tendon forces through inverse kinematics and dynamics calculations. Camera setup tests in various exercise scenarios ensure optimal tracking accuracy. The findings indicate effectiveness of the system in enhancing efficiency in weight training, which provides a method for non-contact training monitoring.**

## I. Introduction

Weight training is an essential component of athletic training across all sports disciplines, crucial for enhancing strength, endurance, and explosiveness in targeted muscle groups. This form of training is universally adopted as a key strategic element in athletic conditioning. However, weight training, especially with heavy weights, poses a significant risk of injury to athletes. While coaches play a vital role in guiding athletes through various exercises, their ability to assist becomes limited as the weight increases, particularly for specialized training objectives. As a result, there is a need for a weight training monitoring system. Such a system would be specifically tailored to meet the requirements of athletes, ensuring safer and more effective training sessions. It should not only help in preventing injuries by ensuring proper form and technique but also aid in optimizing the training regimen to match the athlete's individual physical capabilities and goals[1].

Current research often employs IoT devices or wearable sensors for data collection and analysis in fitness progress assessments [2, 3]. However, these methods lack direct guidance during exercises. In contrast, camera-based methods enable real-time monitoring of human body posture. Optical cameras paired with well-designed markers accurately track joint movements [4], feeding data into programs for motion quality feedback. Wearable sensors like Electrical Impedance Tomography (EIT) have also been used for muscle activation simulation [5], but they may impede users during weight training. Although effective, optical camera systems with markers can be impractical due to their complex setup, need for specific suits with markers, high costs, and site requirements with over four cameras, limiting their everyday usability.

Given the focus on weight training, the study avoids wearable sensors for muscle data collection. Instead, it explores using OpenSim [6], a muscle simulation platform, for deriving muscle fiber and tendon forces through iterative calculations. This approach depends heavily on model definitions and body structure parameters. Previous studies have successfully combined optical cameras with markers and OpenSim for real-time gait analysis [7], encompassing a comprehensive workflow from camera setup to OpenSim modeling. There is also emerging research using depth cameras and skeleton recognition algorithms [8, 9] for identifying human body joints as virtual markers, though these lack depth in fitness training monitoring and upper limb analysis.

Addressing the specific requirements of weight training, this paper presents a novel assessment system utilizing the Orbbec Femto Mega depth camera. This innovative system employs skeleton recognition technology to process and convert data into joint trajectories, which are then inputted into an advanced upper limb model within OpenSim. This detailed model encompasses comprehensive representations of upper limb muscles and joint mechanics, enabling the execution of inverse kinematics and dynamics analyses. Consequently, it allows for the precise evaluation and visualization of muscle fiber and tendon forces. Additionally, the study undertakes a series of camera tests to identify the most effective setup configurations for conventional weight training environments. By conducting simulations within OpenSim, the research assesses the practicality of markerless simulations and enhances the overall functionality of the system, paving the way for more personalized and safer weight training practices.

## II. Methodology

### A. System Architecture

In the realm of weight training, real-time data acquisition is critical for validating the accuracy of an athlete's movements. Key metrics such as the angles and angular velocities of joints serve as essential indicators for assessing training effectiveness. This data informs athletes about maintaining correct posture, appropriate exercise range, and applying the correct muscle force. During exercises conducted on specialized training equipment, the seamless transmission of data is integral to the system's overall functionality.

As illustrated in Figure 1(a), this system's architecture captures the athlete's motion at a rate of 30Hz. Each captured frame undergoes real-time processing to detect the skeleton using Orbbec pose, which is a 3D skeleton detection algorithm.

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The skeleton detection algorithm identity 21 points of human body and output its 3D position, confidence level and sheltering situation. The monitoring system reads out the 3D position of detected body and calculate the joint angle between nearby joints. Then, the data is transmitted to a training machine via the UDP protocol as shown in Figure 1(b). The training machine help a human subject achieve isometric muscle exercise through adjust the joint angle get from the depth camera.

The three-dimensional coordinate data of 21 joints is acquired through the depth camera. Trajectory of these joints is primarily derived from the joint angle, which is computed in real-time as the angle between 3D vectors, according to the following equation. Here, θ represents the angle at Joint A, Joints B and Joint C are adjacent to Joint A.

$$\theta = \arccos\left(\frac{\overrightarrow{AB}\cdot\overrightarrow{AC}}{\|AB\|\cdot\|AC\|}\right) \qquad (1)$$

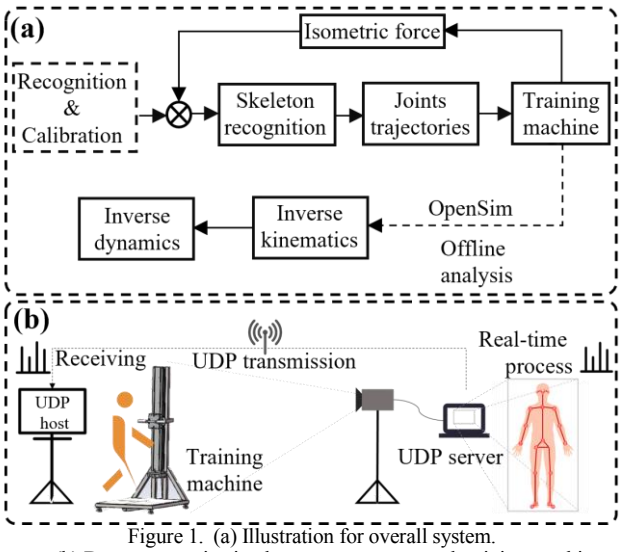


Figure 1. (a) Illustration for overall system.
(b) Data communication between computer and training machine.

## B. Analyze Method

OpenSim is used in offline analysis. OpenSim [6] is an open-source biomechanical analysis tool developed by Stanford University. It offers various human body models with attached virtual markers, each associated with a skeleton joint-bone. By mapping joints to markers within these models, motion analysis can be performed. Through visualizations of the analysis and plots of muscle and skeletal involvement, a comprehensive assessment of the subjects' exercise motions can be obtained.

The recorded skeleton data is processed to the trace file in a specific format. And further inverse kinematic analysis is conducted in OpenSim. A Simbody model [10] with more degrees of freedom is selected to have a more accurate simulation. Before the analysis, the coordinate system of virtual markers on Simbody need to be aligned with the joint detected by depth camera as shown in Figure 2.

The concept of inverse kinematics [6] involves determining the most likely positions of points that were not recorded between nearby position recordings, based on the given sample rate and point coordinates. The trajectory of body joint will be simulated through a weighted least squares method. An iterative process is employed to determine the joint position vector through this optimization approach, using the weights to guide the motion patterns of the joints. The outcome of the inverse kinematics analysis is the angular variation in each specified joint, which is captured in a motion file for further analysis in inverse dynamics.

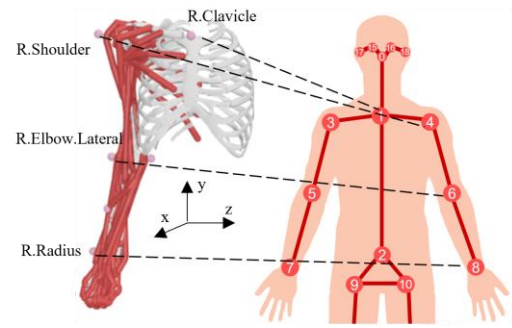


Figure 2. Mapping relation between joints and markers.

Then, inverse dynamics analysis enables the calculation of fiber force and torque as given the trajectory of the joints. Thus, the utilization and activation of muscle could be analyzed through inverse dynamics tool.

# III. Experimental Result

## A. Camera Preparation

Accurate detection and stable tracking of body are crucial for weight training. The camera position needs to ensure that the detected skeleton does not be prone to drifting or loss of tracking throughout entire process. Additionally, the camera must be positioned appropriately to avoid collision or hindrance to movement. Hence, we conduct camera experiments in gym environment to determine the optimal camera setup positions for various exercises.

During the site tests, we focus on four common exercise motions: deadlift, squat, bench press, and bicep curls. For the deadlift, as illustrated in Figure 3, we conducted camera tests from three different sides, while front-side and a 45° side view both exhibits skeletal misalignment. A pure side view results in lost tracking of joints. However, a front view with a 45° elevation angle of the camera provides stable and accurate skeleton recognition and tracking during the deadlift. In the bench press test, we initially use a 45° side view accompanied by a 26° pitch angle position to enable full-body detection. During this motion, we observe that the joints of the upper limbs are prone to loss of tracking, which also occurs with a front view setup. To obtain a better view of individuals lying down, we employ a front view with a 45° pitch angle of the camera. For squats and bicep curls, front views and pure side views respectively provides suitable angles without requiring specific adjustments.

These camera tests allow us to identify the most suitable camera setup positions for each exercise, ensuring accurate joint detection and tracking throughout the movements. The optimal camera positions are determined based on considerations of skeletal misconnections, tracking consistency, and overall visibility of the exercise actions.



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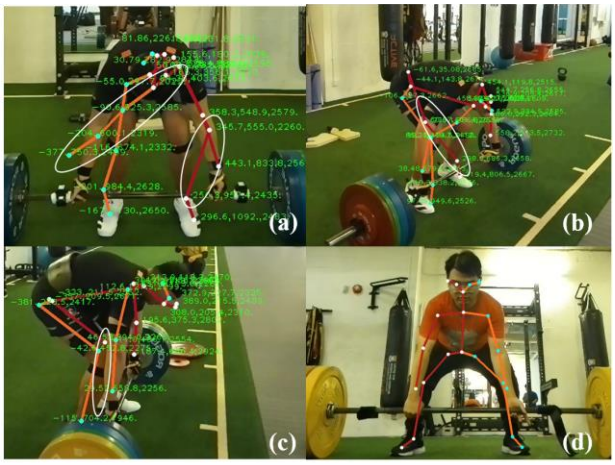


Figure 3. (a) Horizontal front view. (b) 45° side view. (c) Pure side view. (d) Front view with 45° elevation angle.

Figure 4. (a) Side view of bench press, (b) Front view with downward 45° pitch of bench press. (c) Front view of squat. (d) Side view of bicep curl.

### B. *Simulation Result of OpenSim*

To enhance simulation accuracy, we chose a model [10] that offers increased degrees of freedom for each joint. During the conversion process, the primary objective is to map the joints to the model's markers. During the data conversion phase, our primary aim is to accurately align the physical joints of the subjects with the corresponding markers defined within the model. This intricate process involves the translation of joint indices into the specific marker names as stipulated by the model's architect. Subsequently, these marker trajectories are meticulously recorded into a file format that is compatible with OpenSim, thereby facilitating the seamless recognition and analysis of the data within the simulation environment. It should be note that, the selection of markers must be aligned with the bones involved in the movement. For example, when examining elbow flexion, the primary focus should be on the movement of the humerus and ulna, as these are the bones directly involved in the action. Conversely, the movements of the scapula and wrist are less pertinent to this specific analysis and can be omitted without compromising the integrity of the simulation.

Our research places a significant emphasis on the study of arm flexion, employing both inverse kinematics and dynamics to gain insights into the mechanics of this movement. The congruency between the simulation outcomes and the actual data on arm movements is visually represented in Figure 5, displaying the congruence between simulation outputs and actual arm movement data. Furthermore, the results from inverse dynamics, as shown in Figure 6 (a) and (b), provide a detailed examination of the variations in torque and muscle fiber force throughout the movement cycle.

These findings are in alignment with extant studies on the torque produced by the bicep long muscle fiber during elbow flexion [11]. Such studies have typically utilized Biodex systems and Electromyography (EMG) sensors to measure torque, serving as a benchmark for our simulation outcomes. Remarkably, our simulation results (Figure 6 (a)) for the bicep long muscle fiber closely mirror the trends observed in previous research result [11] , validating the efficacy of our system in modeling and analyzing the dynamics of arm motion. The result shows the potential of our system to serve as a tool in the biomechanical analysis of movement, offering insights that are both accurate and relevant to the field of exercise science.

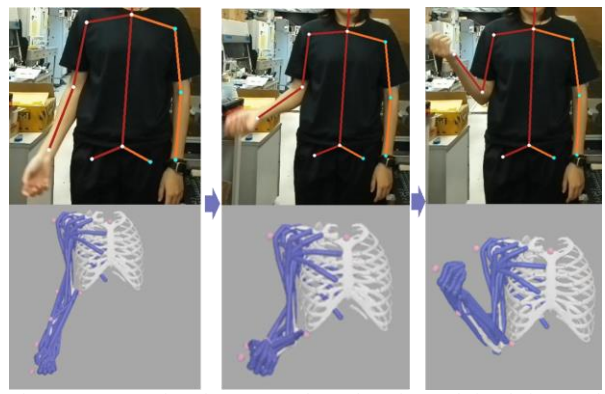

Figure 5. Comparison between real-word motion and simulation result.

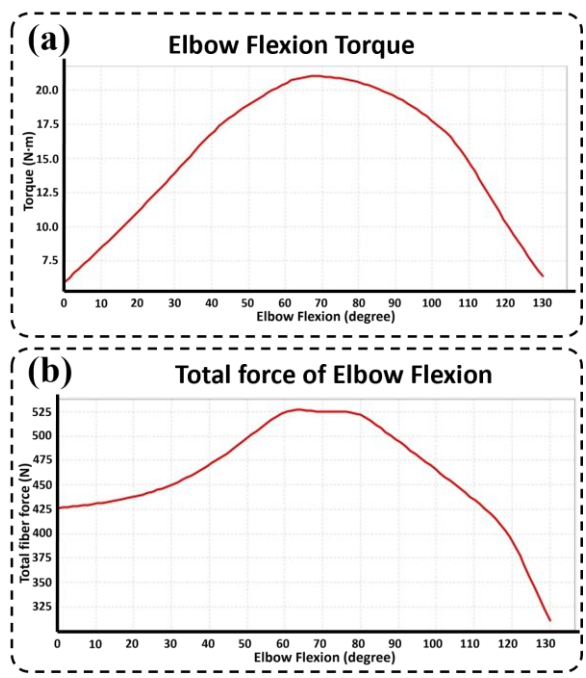


Figure 6. (a) Simulation result of torque of BCI long fiber. (b) Simulation result of total force of BCI long fiber.



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## IV. DISCUSSION

This study explores the optimization of depth camera settings with the objective of refining motion detection, assessment, and simulation in exercise contexts. The research aims to pinpoint the most effective camera configuration tailored for real-life training scenarios. The findings of these tests reduce tracking and enhancing the overall efficacy of exercise monitoring. Furthermore, the utilization of OpenSim for offline simulation has provided insights into muscle activation patterns, offering an understanding of the biomechanical aspects of exercise movements.

Despite these advancements, the study acknowledges certain limitations. One of the issues encountered is the drifting observed in skeleton tracking at the commencement of monitoring. This affects the reliability of the data collected. To address this, future research will concentrate on developing a real-time optimized filter aimed at minimizing the inaccuracies in skeleton data. Kalman filter and Butterworth filter will be considered in real-time processing [12-14]. Furthermore, the methodology will be evaluated to ensure both its effectiveness in reducing noise and its ability to maintain accurate tracking.

Another promising avenue for future research involves the implementation of a multiple-camera system. By integrating data from several cameras, it is expected that the accuracy of motion capture methods [15] will be further improved. This multi-faceted approach will likely offer a more comprehensive view of the subject's movements, thereby overcoming some of the limitations associated with single-camera setups.

Moreover, the study aims to extend its research scope to include fatigue detection during weight training exercises[16, 17]. Robust neural network should be learned [18] and developed, and considering the real-time process capability, efficient architecture [19] is needed as well. Recognizing the critical link between the velocity loss of body joints and muscle fatigue, the development of a neural network model is planned. This model will be designed to accurately detect when a subject may require assistance during their training sessions, based on the signs of fatigue.

## V. CONCLUSION

In conclusion, this research highlights the role of depth camera settings in enhancing the detection and simulation of motion within exercise environments. By addressing the current limitations and proposing future directions for improvement, this study contributes insights and potential solutions to the challenges of monitoring and assessing human exercise movements. The continued exploration and development in this field promise to advance our understanding and capabilities in sport science, leading to more effective and personalized training regimes.

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